# Hints: Linear algebra – 6. Linear algebra in combinatorics

Class work

*The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.*

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## Problem 1

**Hint 1.** Oddtown: over $\mathbb{F}_2$ the Gram matrix $(\langle 1_{A_i}, 1_{A_j}\rangle)$ is the identity. So the vectors $1_{A_i} \in \mathbb{F}_2^n$ are independent. The singletons show sharpness.

**Hint 2.** Eventown: now every inner product, including $\langle v, v\rangle$, is 0. So $W = \operatorname{span}\{1_{A_i}\}$ satisfies $W \subseteq W^\perp$, and $\dim W + \dim W^\perp = n$ gives $\dim W \leqslant \lfloor n/2 \rfloor$. The family lies inside $W$, so $m \leqslant |W|$. Sharp: all unions of the pairs $\{1,2\}, \{3,4\}, \ldots$.

**Hint 3.** The difference: in Oddtown the vectors are independent, so you count dimensions. Eventown families can be closed under symmetric difference, i.e. be a whole subspace, so you count the elements of a subspace.

## Problem 2

**Hint 1.** The answer is $2n$ (take $\pm e_i$). For the bound, put $M = \sum_i v_i v_i^T$ and compute $\operatorname{tr} M$, $\operatorname{tr} M^2$, $\operatorname{tr} M^3$ in terms of the inner products.

**Hint 2.** Let $S = \sum_{i \neq j} \langle v_i, v_j\rangle^2$. Then $\operatorname{tr} M = m$ and $\operatorname{tr} M^2 = m + S$. In $\operatorname{tr} M^3 = \sum \langle v_i, v_j\rangle\langle v_j, v_k\rangle\langle v_k, v_i\rangle$, terms with $i, j, k$ distinct vanish by hypothesis, so $\operatorname{tr} M^3 = m + 3S$.

**Hint 3.** Apply Cauchy–Schwarz to the eigenvalues $\mu_l \geqslant 0$ of $M$. First $(\sum \mu^2)^2 \leqslant \sum \mu \cdot \sum \mu^3$ gives $S \leqslant m$. Then $(\sum \mu)^2 \leqslant n \sum \mu^2$ gives $m^2 \leqslant n(m+S) \leqslant 2nm$.

## Problem 3

**Hint 1.** Small cases: the minimum is 0 for $n = 1$ and 2 for $n = 2$. For $n \geqslant 3$, try to write $a_{ij}$ as a sum of few products $f(i)g(j)$ vanishing on the diagonal: $(i-j)^2 = i^2 - 2ij + j^2$ has rank $\leqslant 3$.

**Hint 2.** Rank 2 is impossible for $n \geqslant 3$. The first two rows are independent, so row $3 = x\cdot$row $1 + y\cdot$row 2. Columns 1 and 2 force $y > 0$ and $x > 0$, and then column 3 gives $0 = xa_{13} + ya_{23} > 0$. Answer: 3 for $n \geqslant 3$.

## Problem 4

**Hint 1.** Independence of the ordering: reordering conjugates $M$ by a permutation matrix. Then set up an induction on $n$. Order the subsets as: non-empty subsets of $[n-1]$, then $\{n\}$, then $S \cup \{n\}$ for non-empty $S \subseteq [n-1]$.

**Hint 2.** In this order $M_n = \begin{pmatrix} M_{n-1} & 0 & M_{n-1} \\ 0 & 1 & \mathbf{1}^T \\ M_{n-1} & \mathbf{1} & J \end{pmatrix}$. Subtract the first block row from the third, the first block column from the third, and then the middle row from the last block.

**Hint 3.** You get $\det M_n = \det M_{n-1} \cdot \det(-M_{n-1}) = -(\det M_{n-1})^2$, since $M_{n-1}$ has odd size. With $\det M_1 = 1$: $\det M_n = -1$ for $n \geqslant 2$.

## Problem 5

**Hint 1.** The affine dimension of a point set equals the rank of its centred Gram matrix $-\frac{1}{2}HDH$, where $D$ is the matrix of squared distances and $H = E - \frac{1}{2n}J$. Here $D = b^2(J - E) + (a^2 - b^2)P$, with $P$ the involution swapping $x_i \leftrightarrow y_i$.

**Hint 2.** $P$ commutes with $J$ and $H$. Split $\mathbb{R}^{2n}$ into vectors symmetric under $P$ (dimension $n$, containing $\mathbf{1}$) and antisymmetric ones (dimension $n$). On $\mathbf{1}^\perp$ the Gram matrix acts as $b^2 - \frac{a^2}{2}$ on the symmetric part (dimension $n-1$) and as $\frac{a^2}{2}$ on the antisymmetric part.

**Hint 3.** So the rank is $n$ if $b^2 = a^2/2$ and $2n - 1$ otherwise. But the points span at most $d \leqslant 2n - 2$ dimensions. Hence $b^2 = a^2/2$ and the points lie in an $n$-dimensional affine subspace.